

SYSTEM DEVELOPMENT LIFE CYCLE (SDLC)

Development Life Cycle (SDLC) is a structured process used by organizations to plan, create, test, and deploy an information system. It's a framework that outlines the distinct phases involved in developing software or IT systems, from initial concept to final implementation and maintenance.

Description of System Development Life Cycle

The SDLC provides a systematic and disciplined approach to building high-quality systems. It helps to ensure that systems meet user requirements, are delivered on time and within budget, and are well-maintained after deployment. By breaking down the complex process of system development into manageable stages, SDLC helps organizations to manage risk, improve efficiency, and achieve their strategic objectives.

Stages in System Development Life Cycle

The SDLC is typically divided into several key stages, which may vary slightly depending on the specific methodology used. The common stages are:

1. Preliminary Study
2. Feasibility Study
3. Planning
4. Analysis
5. Design
6. Development/Implementation
6. Testing

7.Deployment/Maintenance

Description of Each Stage of System Development Life Cycle

==>**Preliminary System Study:** It is a brief investigation of the system under consideration and gives a clear picture of what actually the physical system is.

==>**Feasibility Study:** It is basically the test of the proposed system in the light of its workability, meeting user's requirements, effective use of resources and the cost effectiveness.

==>**Planning:** This initial stage involves defining the project scope, objectives, and feasibility. It includes conducting preliminary investigations to understand the problem or opportunity and determining the resources required.

==>**System Analysis:** In this stage, the detailed requirements of the system are gathered and analyzed. This involves understanding user needs, current system limitations, and defining what the new system should do.

==>**System Design:** This stage focuses on creating a blueprint for the system. It involves defining the architecture, modules, interfaces, and data structures of the system based on the analyzed requirements.

==>**Development/Implementation:** This is where the actual system is built. Developers write code, create databases, and assemble hardware based on the design specifications.

==>**Testing:** Once the system is developed, it undergoes rigorous testing to identify and fix any bugs or defects. This ensures that the system functions correctly and meets the specified requirements.

==>**Deployment/Maintenance:** The final stage involves deploying the tested system to the end-users and providing ongoing support and maintenance. This includes bug fixes, updates, and enhancements to keep the system operational and relevant.

ASSIGNMENT

Write a comprehensive note on System Development Life Cycle